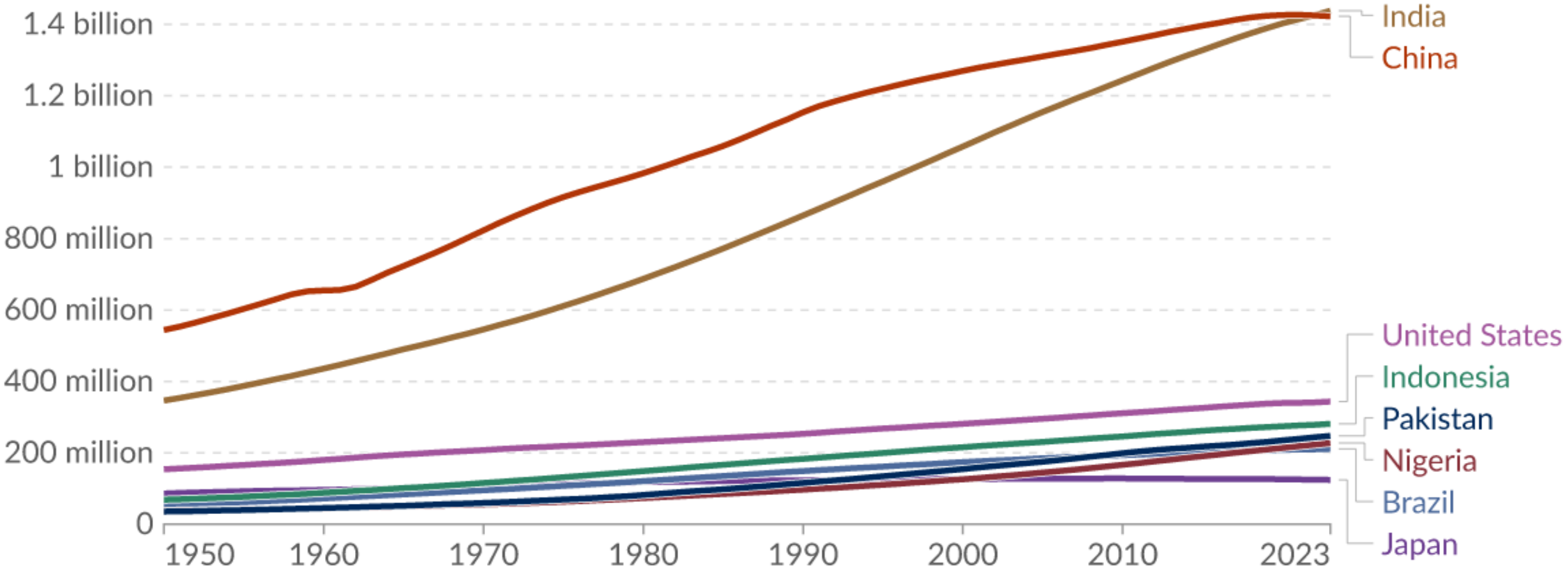
Welcome to SOC 170
Sit on the bottom, not the balcony

Nick Mark

Assistant Professor, Sociology

Demography: The Study of Population



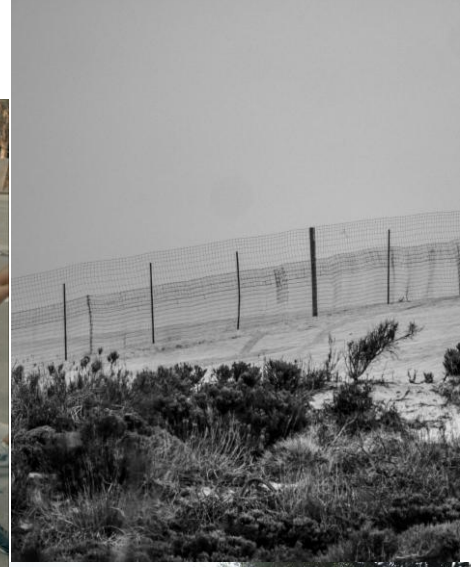
Data source: UN, World Population Prospects (2024)

CC BY

Note: Values as of 1 July of the indicated year.

What topics are we going to cover?

1. Population Frameworks + Change
2. Population Health + Aging
3. Migration



Inequality:

1)Causes of Racial Inequality
- Historical and Contemporary

2)Consequences of Inequality
- Health, Economy, Population

3)Role of Migration

***Foundation**: Understanding the formation of "race" as a concept*

What is on Canvas Lecture Pages?

- Lecture synopsis
- Assigned readings and viewings
- Concepts to know

Lecture 1.1: Introduction to SOC 170, What is Population Thinking? (1/20)

Welcome to our first lecture!

During our first meeting we will discuss the logistics of how the course will flow through the semester, the broad key themes in the material that we will cover, and introduce you to the key concepts of a population and population thinking, and provide an overview of US and global population statistics.

Course Readings:

Course readings should be completed **before** coming to lecture and discussion section. Note that you don't have to read the entire document, only the pages or sections listed in **boldface.** If there are no pages listed in bold, it is expected that you read the entire article, webpage, or chapter. As you read the assigned pages, make sure you understand the key concepts listed in the **Concepts to Know** section of every lecture page. After reading the assigned pages, you should be able to define these terms and explain their significance.

Readings to complete for 1/22: (NOTE: because 1/20 is the first day of class, I set the deadline for these readings for Thursday)

Roser, Max; Hannah Ritchie, and Esteban Ortiz-Espina. (2023) "Population of the World Today" and "How has world population growth changed over time?" *World Population Growth* **[You are only responsible for the sub-sections under the headers "Population of the World Today" and "How has world population growth changed over time?" You can skip all other sections of this text]**

Preston, S.H., P. Heuveline, and M. Guillot. (2001). *Demography: Measuring and modeling population processes* . UK, Blackwell Publishers. **Pp. 1-8, 16-20**

Take the Syllabus Quiz by Tuesday, Jan 12 9:55 a.m.

Concepts to Know:

Population
Count vs Ratio vs Rate
Demographic rates and "Person-years of exposure"
Balancing equation of population
Crude birth rate
Crude death rate
Natural increase
Net migration
Population cartogram

Course Expectations & Assessment

- READ course materials, Attend Lecture and Respond
  - Slides posted 1 week before exam, Within-lecture question:

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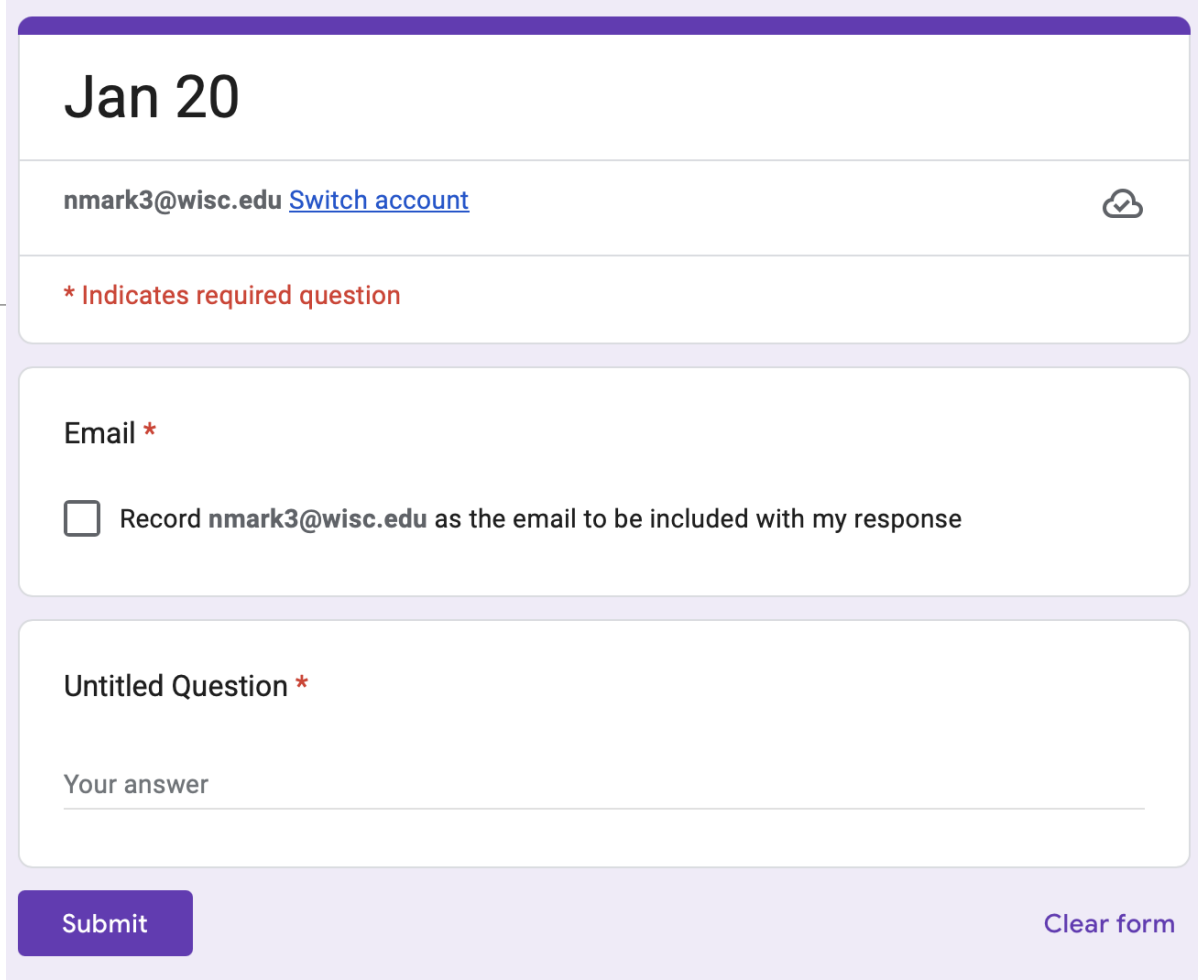
Question:

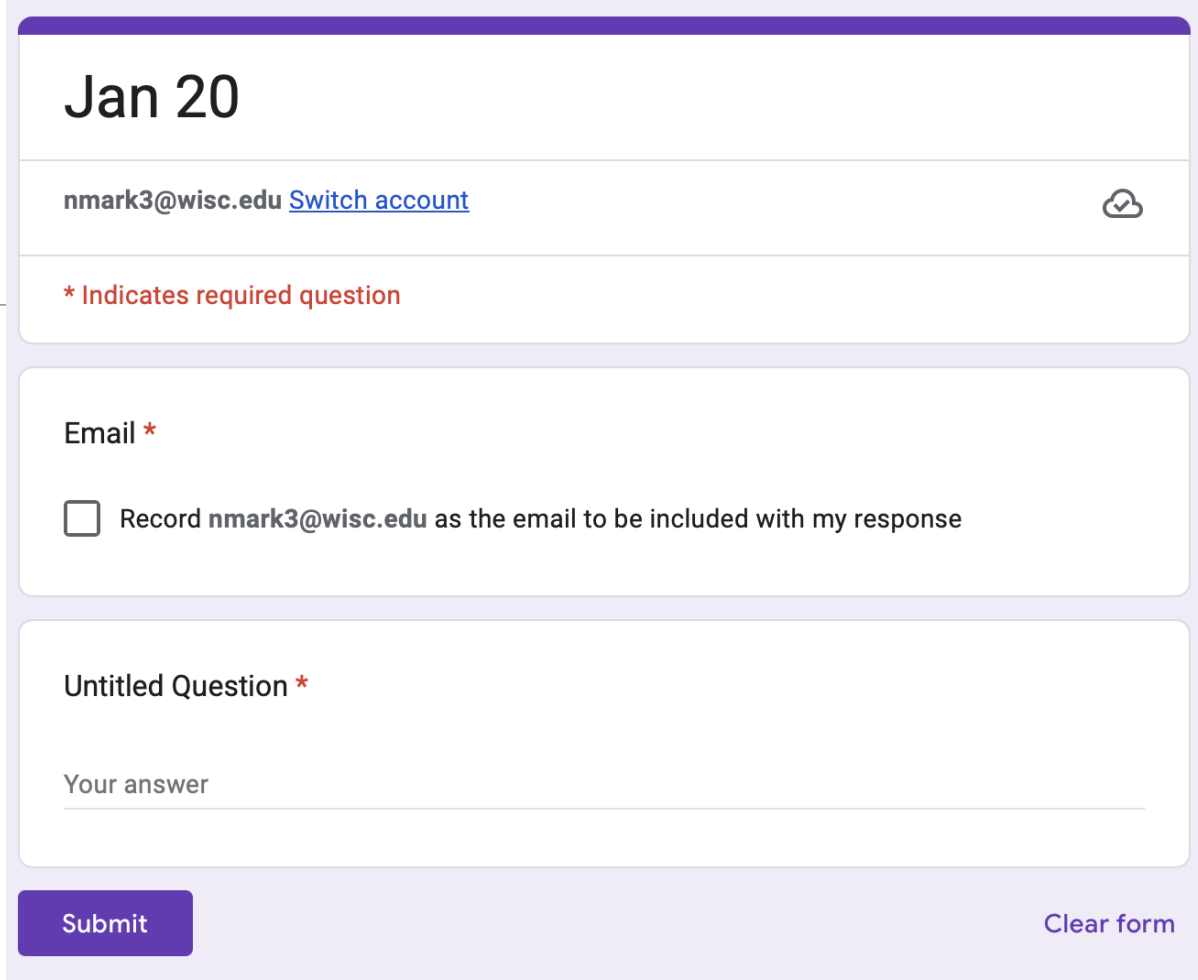
Which do you think is more important for a society in the long run, the size of the population or the rate of change? Explain your answer in 1 sentence.



Question:

Which do you think is more important for a society in the long run, the size of the population or the rate of change? Explain your answer in 1 sentence.

Email Address and Timestamp recorded and used to grade



Course Expectations & Assessment

- READ course materials, Attend Lecture and Respond
  - Slides posted 1 week before exam, Within-lecture question:
- Attend Class
  - Attend class and participate in discussion (20%)
  - Doesn’t just mean talking every class – thoughtful, respectful engagement is key
- Write and apply critical thinking
  - Two writing assignments (450 words each, 20% total)

AI Expectations

AI on its own produces mediocre work.

Simply copy/pasting will result in poor grades.

Do your own thinking for a better grade.

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  - Doesn’t just mean talking every class – thoughtful, respectful engagement is key
- Write and apply critical thinking
  - Two writing assignments (450 words each, 20% total)
- Complete exams
  - Midterm exam (20%), final exam (25%)
- Engage, Reflect, and Respond
  - Extra credit up to 1 percentage point to final grade

Extra Credit: Example

February 3rd, 2026

Jessica Faul (University of Michigan),
"What We Learn by Adding Biological Measurement to an Interdisciplinary, Longitudinal, Social Science Survey: DNA Methylation Research in the Health and Retirement Study."
Co-sponsored by the Department of Population Health Sciences.

1) Attend
2) Take a picture of the speaker
3) Write <500 words:
   1) Summarize the talk.
   2) What is something new that you learned?

Meet your TAs, Hanbi and Bijoyetri

Who should I talk to?

Administrative needs (section changes, for example)

- Ted Babcock, Sociology Admin – ted.babcock@wisc.edu, please cc your TA
- There is an app specifically for section changes – makes it very easy:
- https://sociology.wisc.edu/undergraduate-program/sociology-enrollment-help/#changing-sections

Course Logistics (McBurney accommodations, dates, office hours, etc.)

- Syllabus and Canvas, then your TA (your TA will communicate with me if they don't know)

Course Content (confused about a topic?)

- Canvas, then discussion materials, then classmates, then your TA, then Nick (after class or office hours)

Discussion Section Absence

- You TA, ahead of your absence unless it is an emergency

What do I need to do now?

1. McBurney Accommodations – complete by second week of class

2. Complete the Syllabus Quiz by Tuesday, Jan 27, 9:55 a.m.

3. Before Tuesday's lecture:
   Do the assigned readings for lecture 1.2 (only read the sections in boldface)
   Complete the reading quiz before 9:55am, Sept 10.

4. Mark our exams in your calendar! Both in class.
   Exam 1 - Thursday March 3rd 9:55am – 10:45am
   Exam 2 - Thursday April 30the 9:55am – 10:45am

What is Population Thinking?

What is Population Thinking?

- Population Basics
- Population Equations
- Demographic Rates

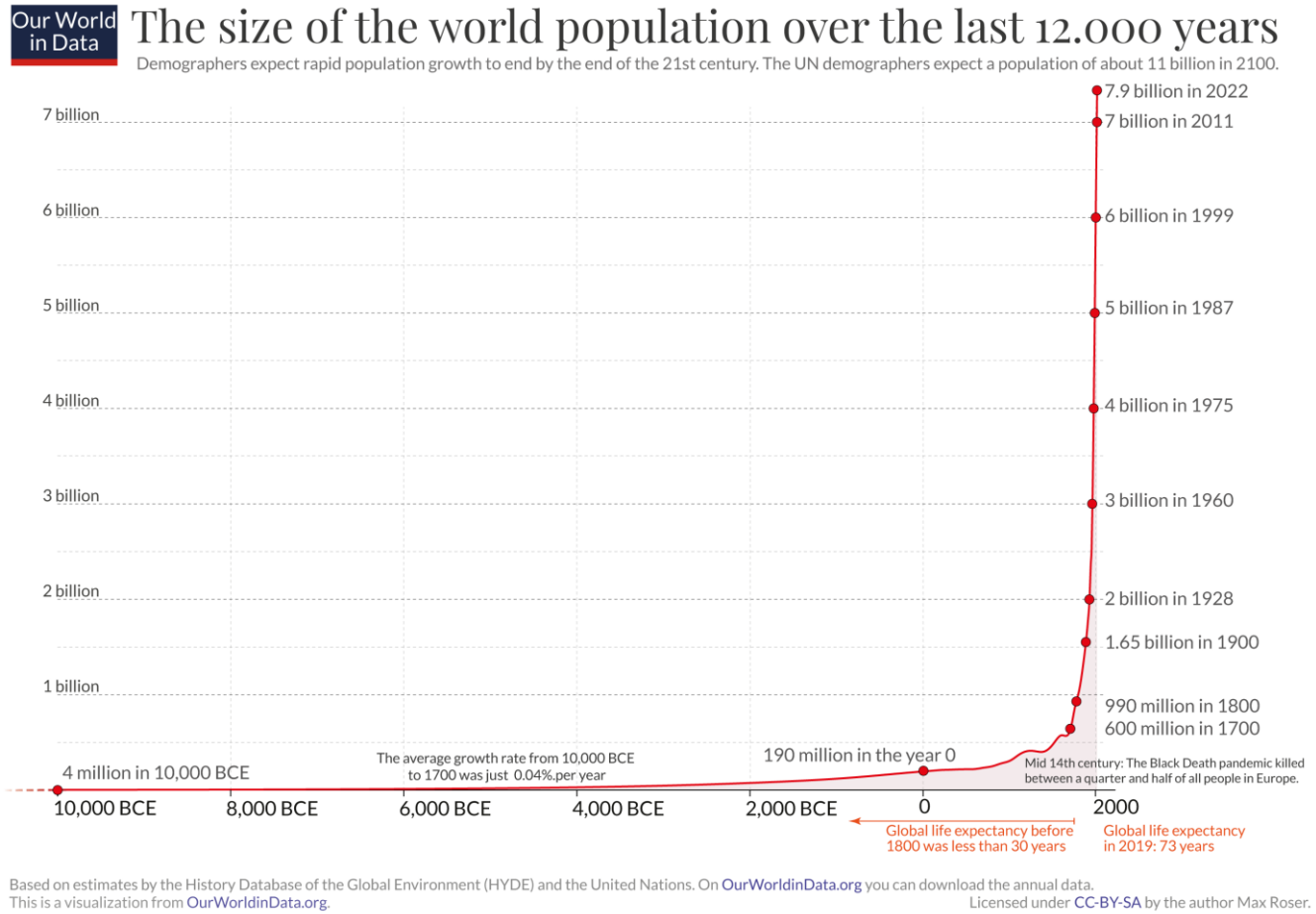
What is a population?

- *Empirical:* Collection of people alive at a specified point in time who meet certain criteria
  - Population of Wisconsin on July 1, 2021 (5,895,908 people)
  - Population of UW students on September 5, 2023 (50,662)

- *Conceptual:* Collectivity of people that persists over time even though its members are always changing
  - The “population of Wisconsin”

What’s important about a population?

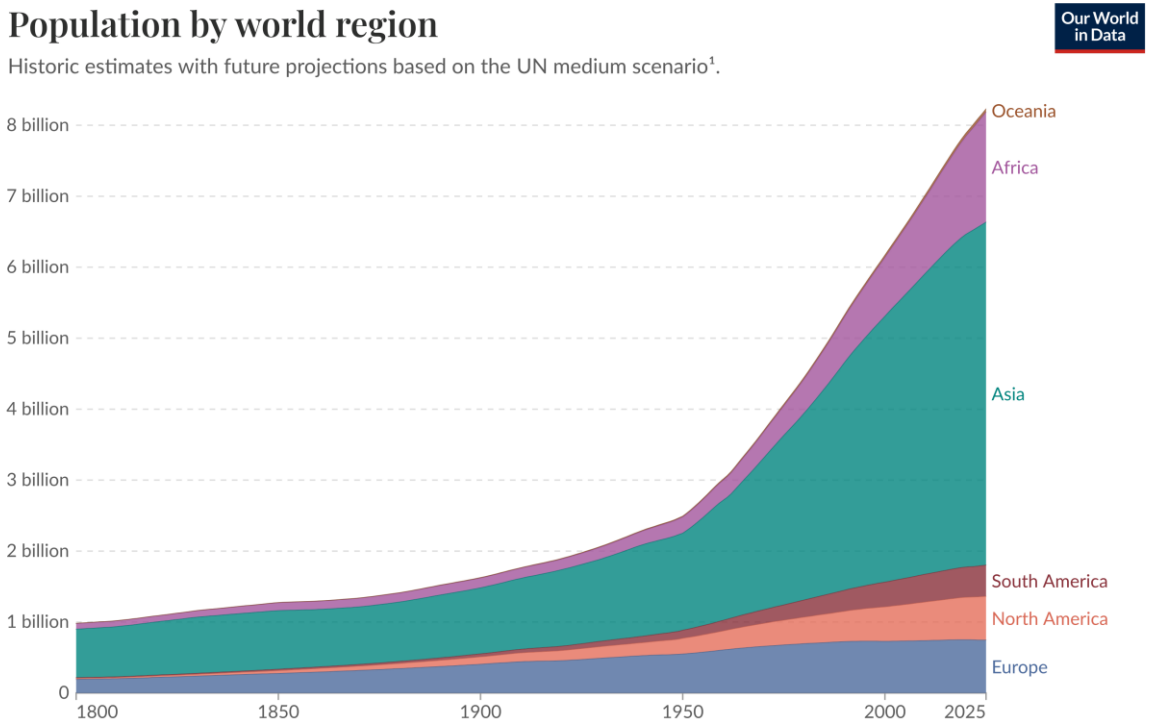
1. Size



What’s important about a population?

1. Size

2. Composition



1. **United Nations projection scenarios** The United Nations' World Population Prospects include a range of projected scenarios for population change. These scenarios rely on different assumptions for fertility, mortality, and migration patterns to project different demographic futures.
Read more on population.un.org

What's important about a population?

1. Size
2. Composition
3. Distribution

population cartogram:
size of shapes determined by population

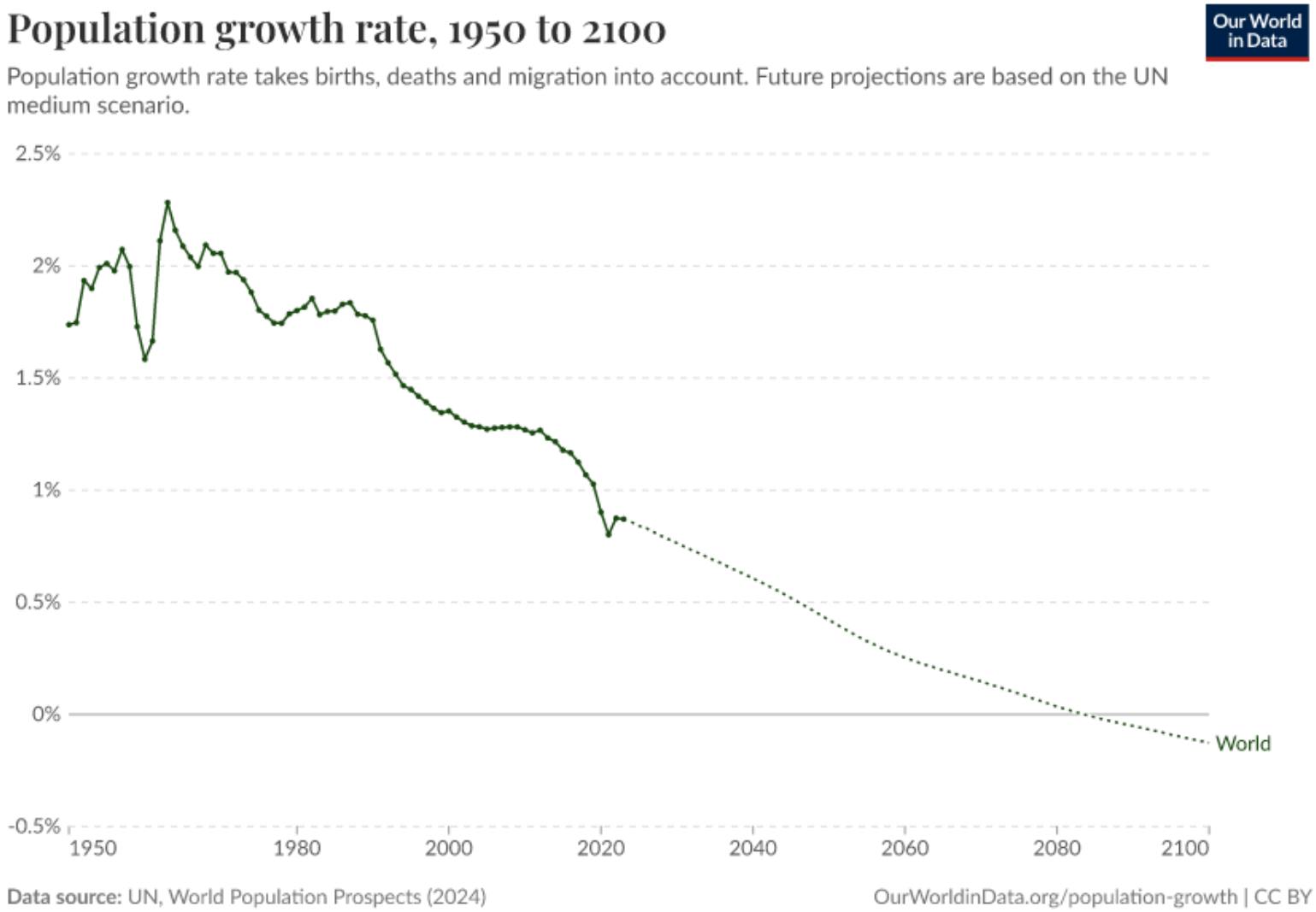
What’s important about a population?

1. Size
2. Composition
3. Distribution

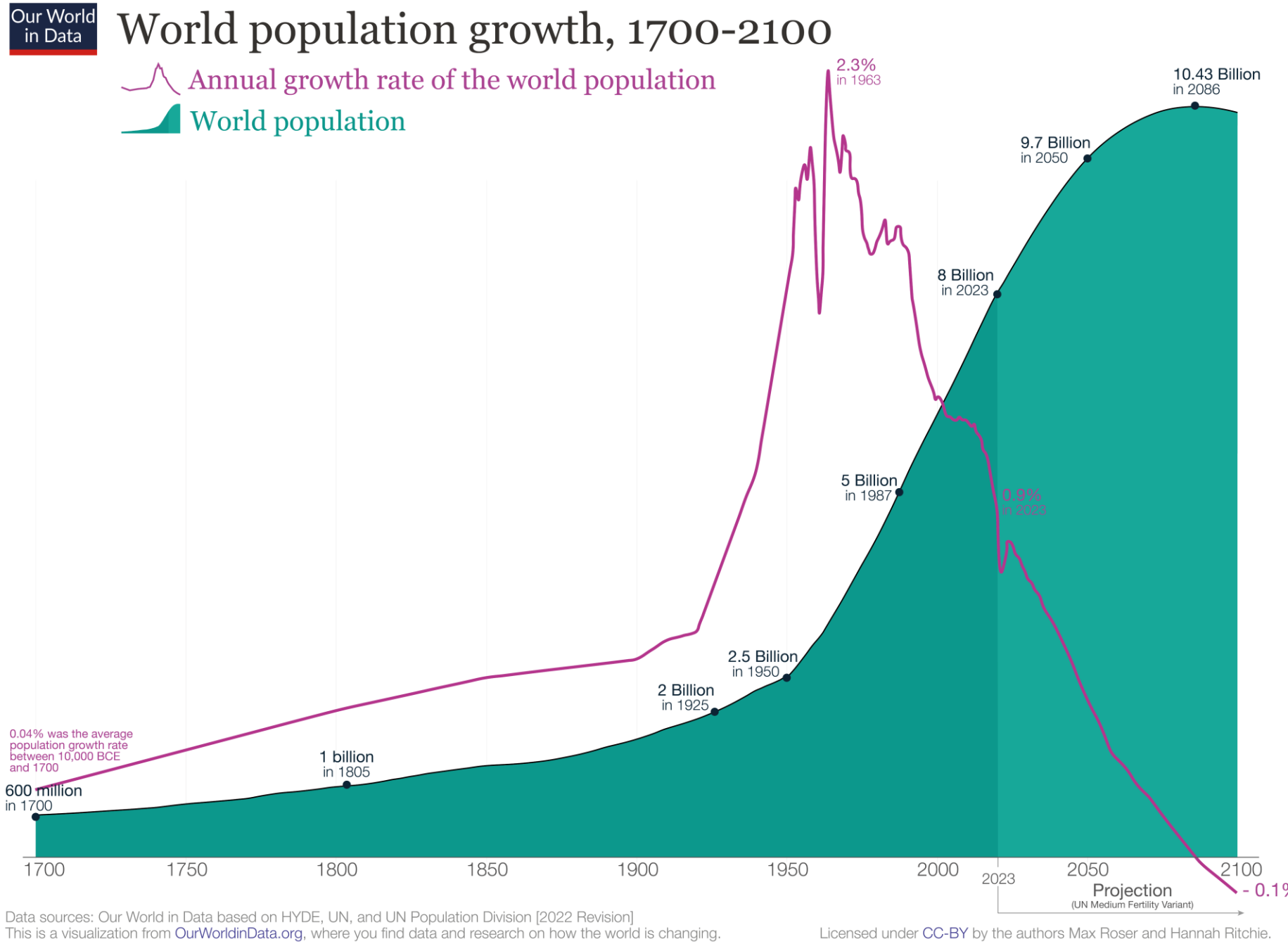
Source: Karim Douieb - https://observablehq.com/@karimdouieb/try-to-impeach-this-challenge-accepted

What’s important about a population?

1. Size
2. Composition
3. Distribution
4. Dynamics



Contrast: Growth vs. Growth RATE



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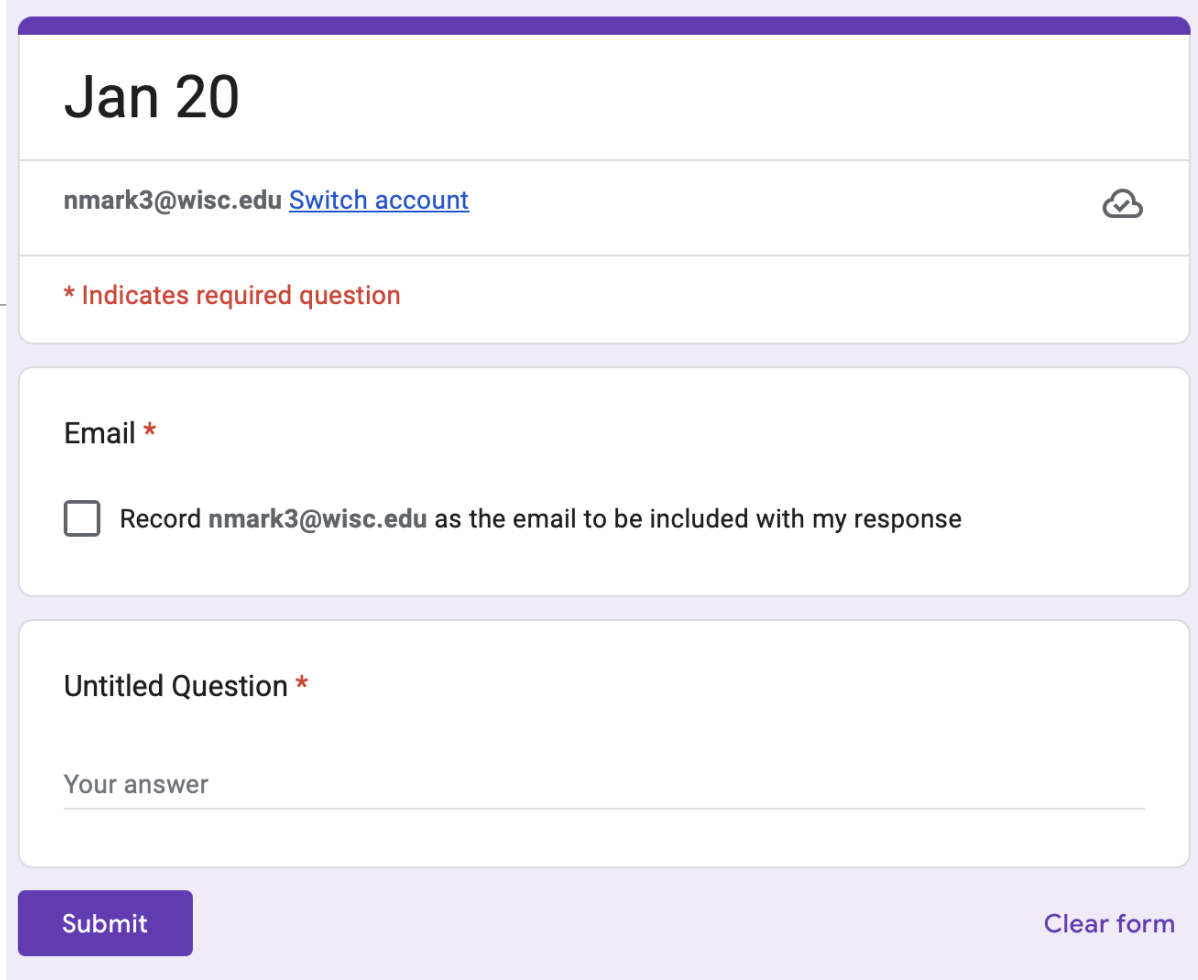

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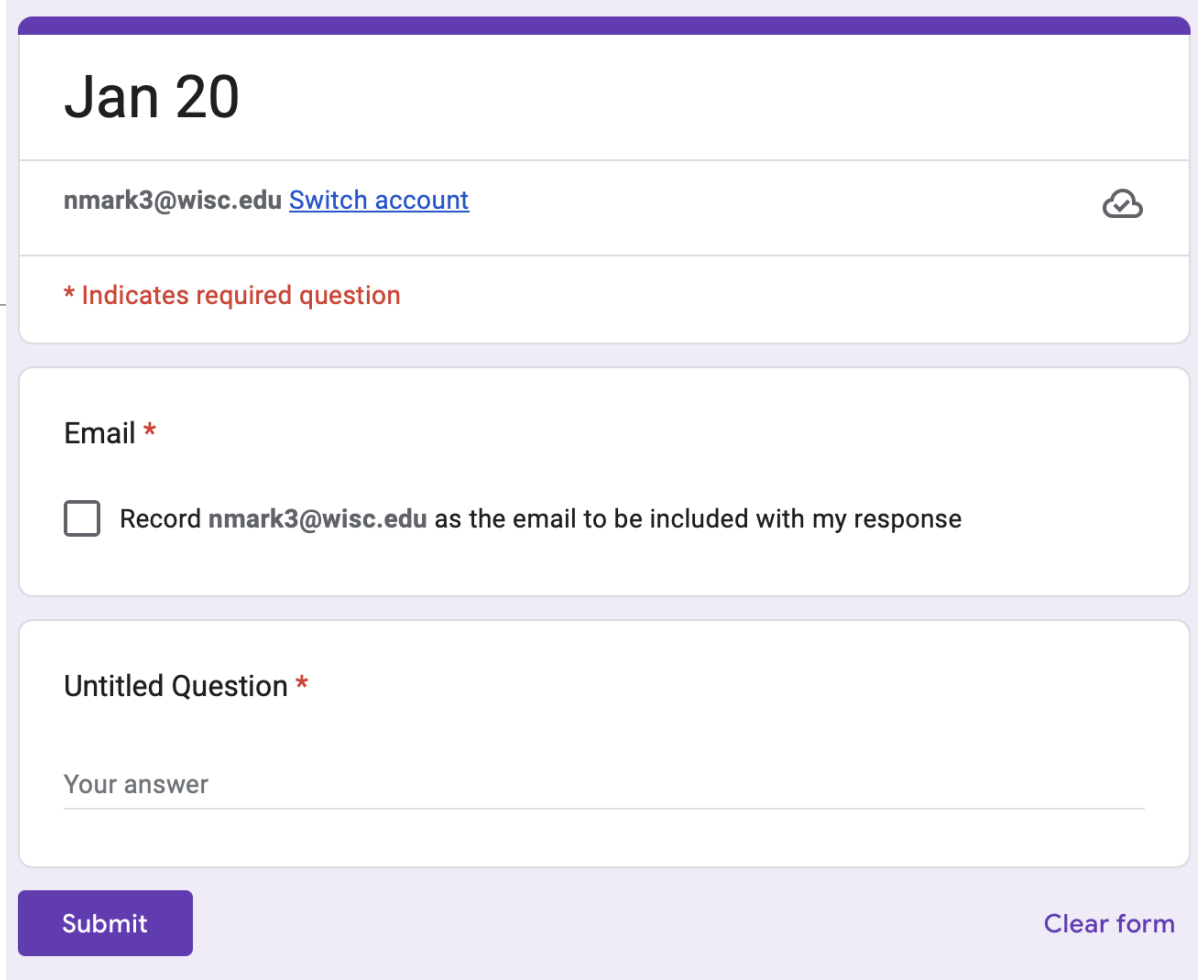
https://tinyurl.com/jan20soc170



Growth Balance Equation

$$\text{Population}_{T2} = \text{Population}_{T1} + \boxed{B - D} + \boxed{I - O} + e$$

Natural Increase/decrease

Net Migration

Demographic Rates

What event are we measuring? Over what time period?

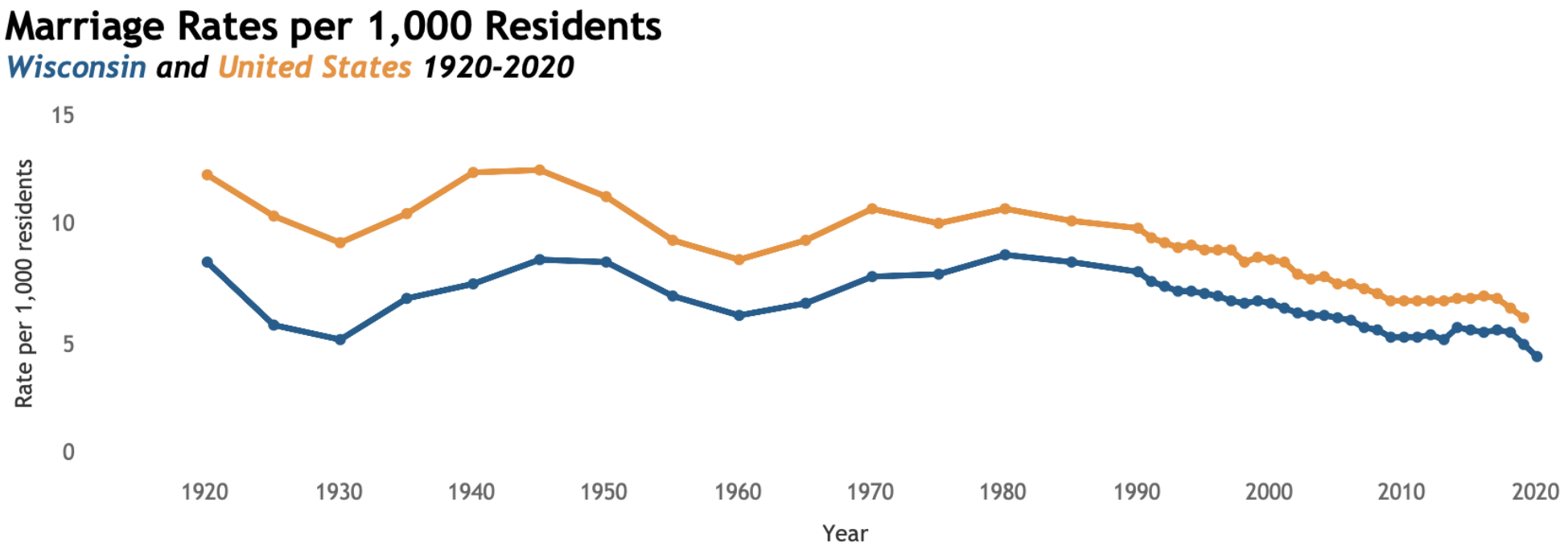
→ This is the **numerator of the rate**

Who is in the population?

→ This is the **denominator of the rate**

$$\frac{\textbf{\textit{Number of Events in Time Period}}}{\textbf{\textit{Population in Time Period}}}$$

Calculating Marriage Rate



Marriage Rate = number of marriages/number of residents

Rate of Population Growth

$$\frac{P_2 - P_1}{P_1} = \frac{\text{Births}}{P_1} - \frac{\text{Deaths}}{P_1} + \frac{\text{Net Migration}}{P_1}$$

Rate of population growth

Birth rate

Death rate

Rate of net migration

Better rates:

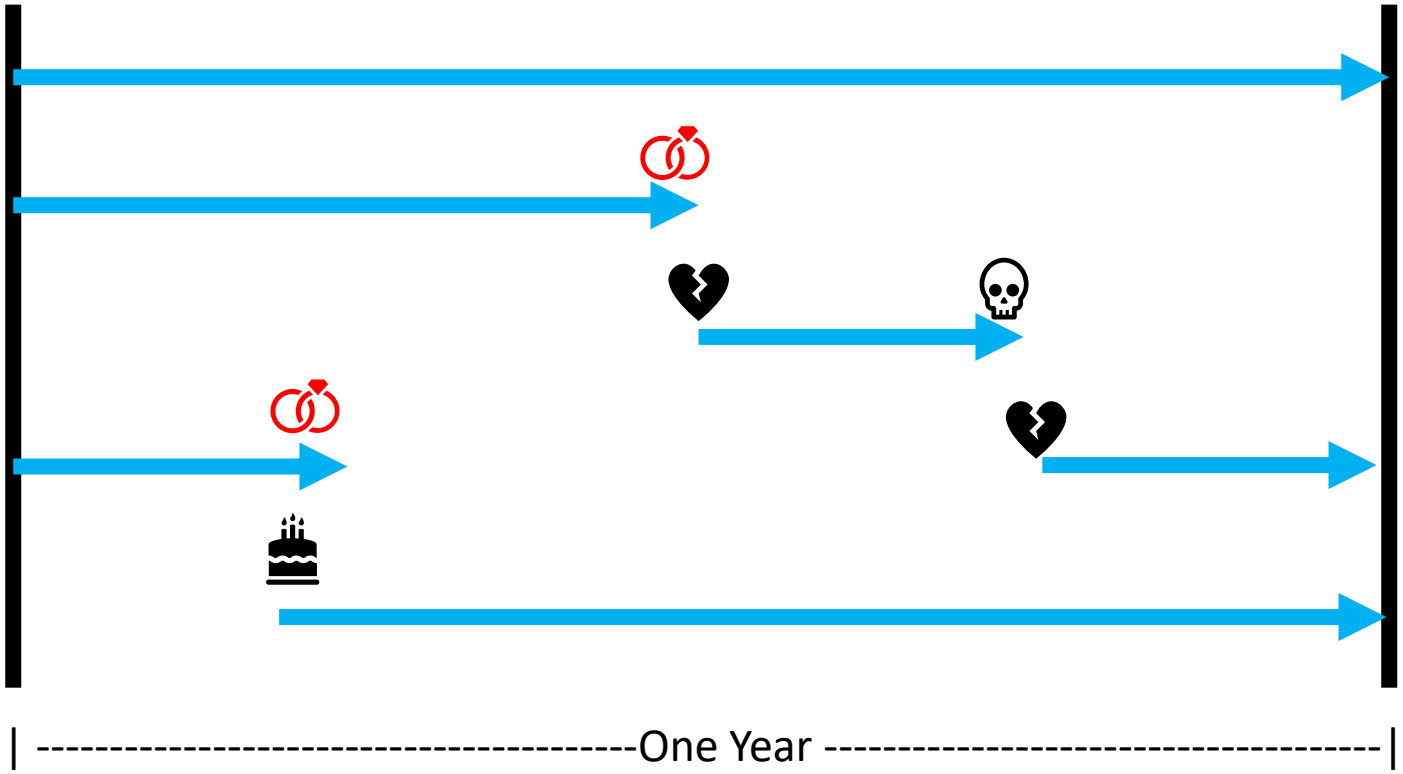
In a *crude rate* the denominator is the total number of people.

But we can define the denominator more precisely!

Who is *at risk* of experiencing an event over this time period?

- How many people are in the population who could actually experience the event? (ex: Can babies get married?)
- What portion of the time period do they spend in the population before they experience the event?

Calculating Crude Marriage Rate



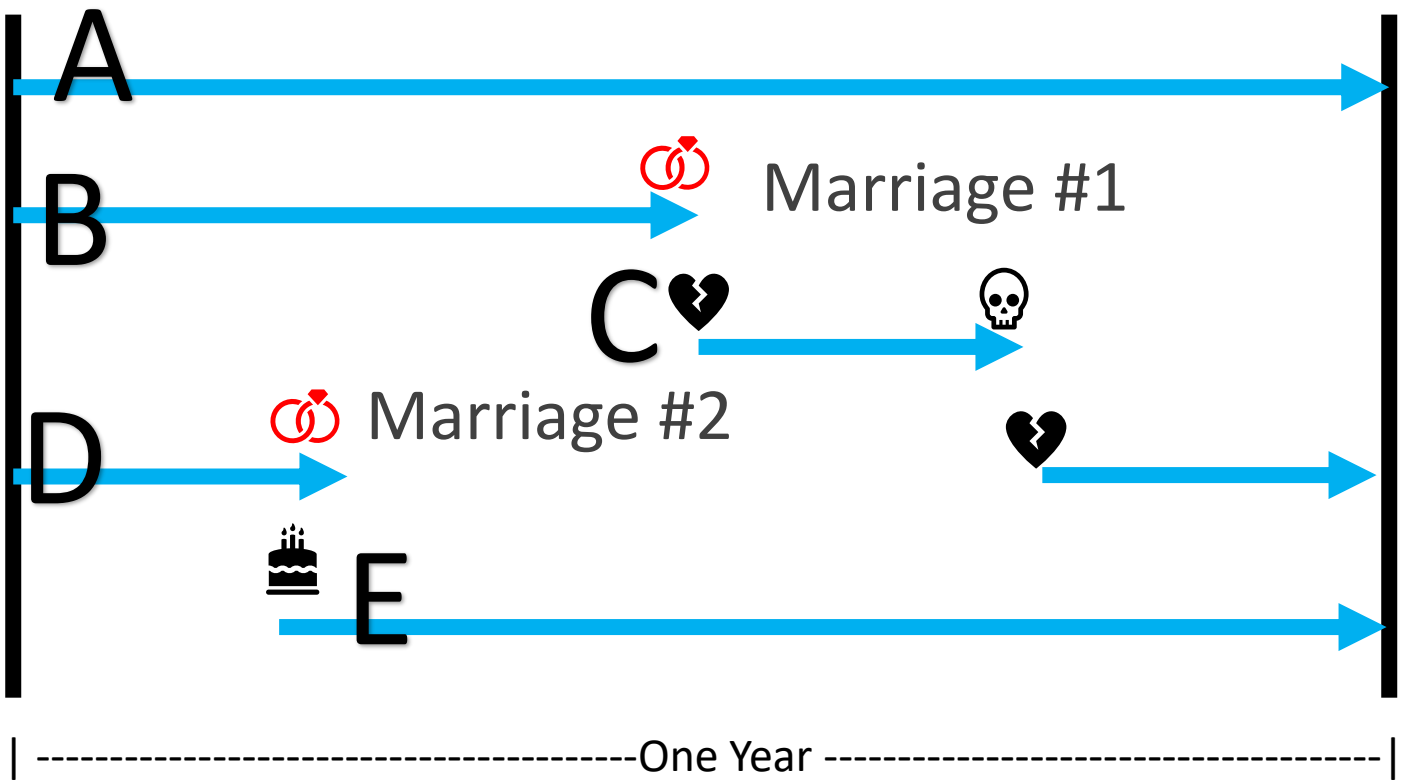
Demographic Event:
Marriage

Number of events:

Population Size:

Crude Marriage Rate:

Calculating Crude Marriage Rate



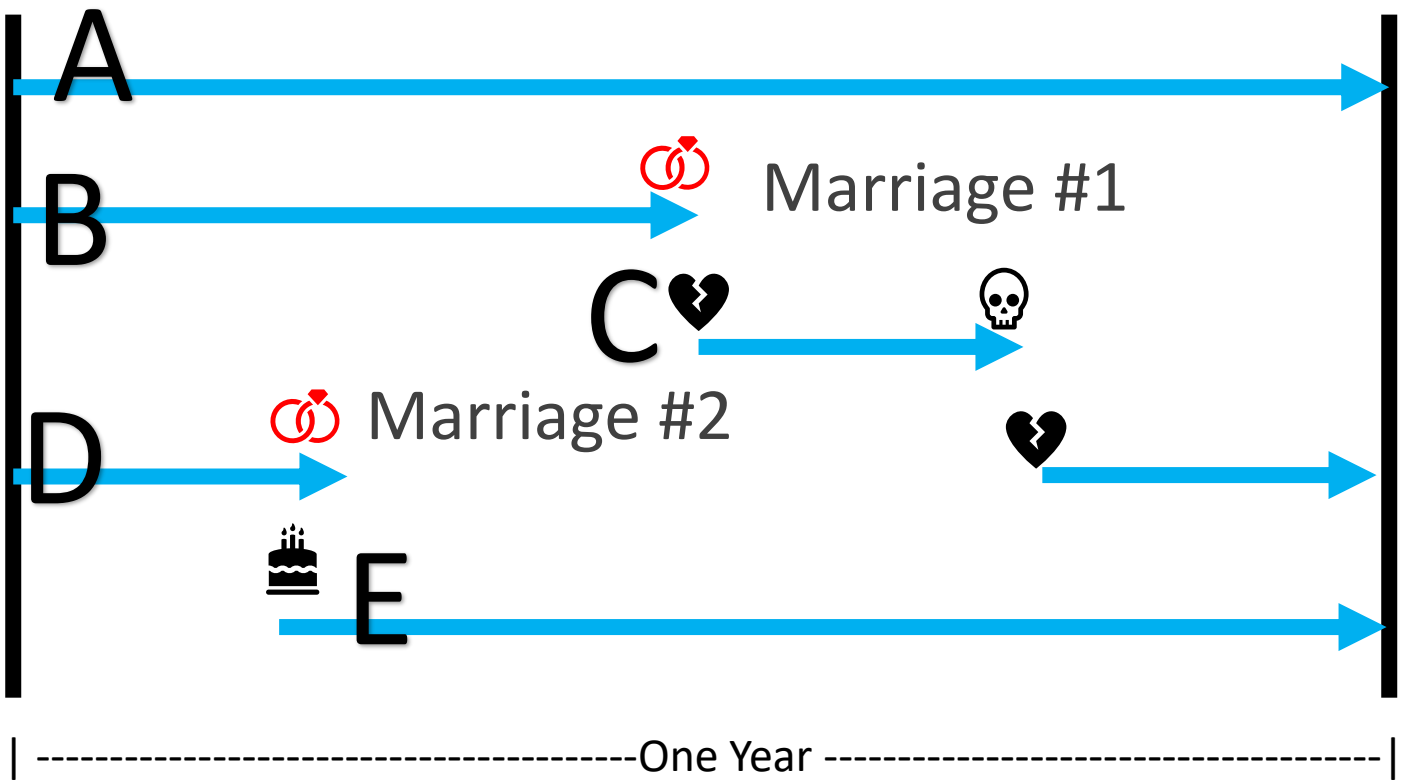
Demographic Event:
Marriage

Number of events:

Population Size:

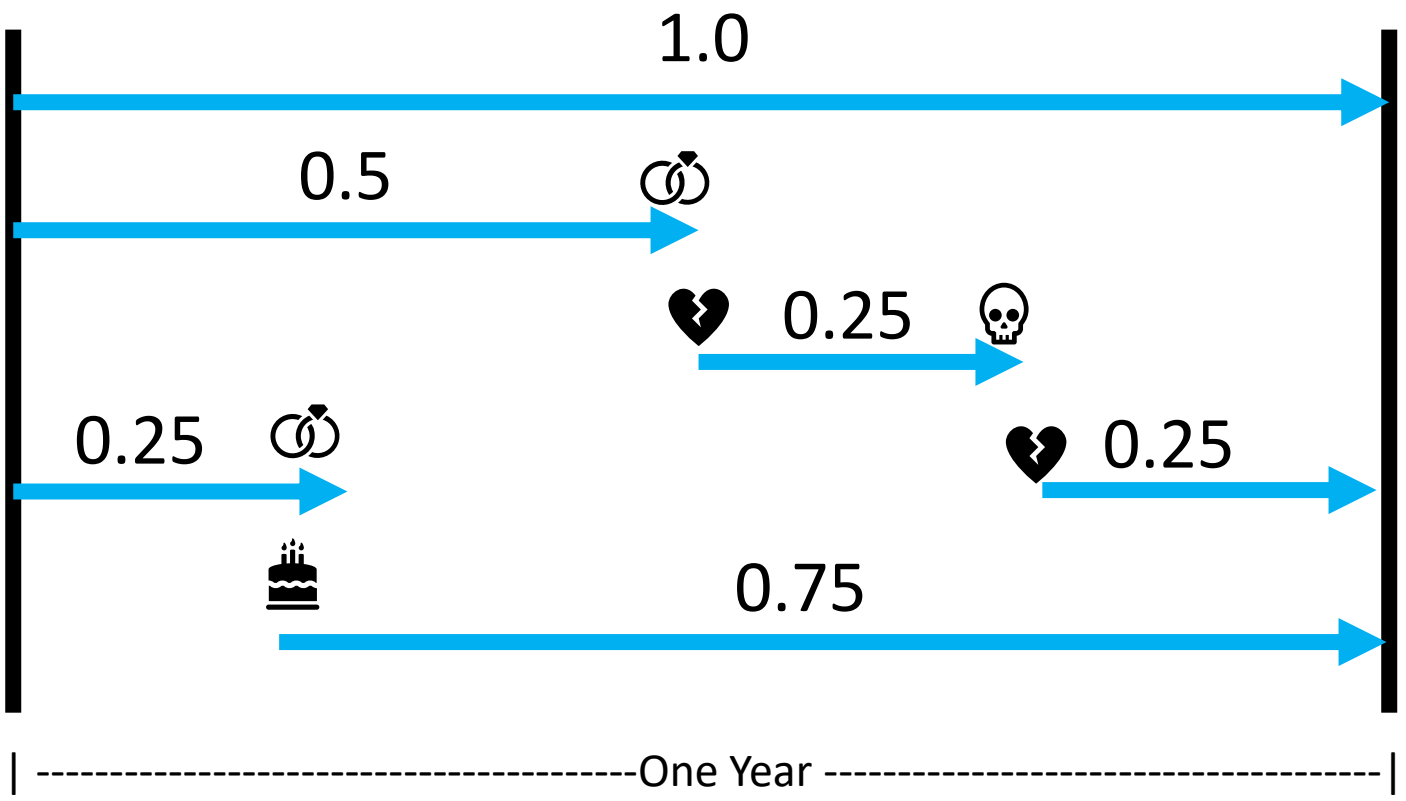
Crude Marriage Rate:

Calculating Crude Marriage Rate



Demographic Event:
Marriage
Number of events:
2
Population Size:
5
Crude Marriage Rate:
2/5

Marriage Rate=
Number of events/number of person-years at risk



Number of events = 2

Number of person-years at risk =
1 + .5 + .25 + .25 +.25 + .75 = 3

Rate = 2 marriages / 3 person-years

Different Rates Can Tell Different Stories

Crude mortality rate

(# deaths/# people)

Japan: 11.7

USA: 8.4

Different Rates Can Tell Different Stories

Crude mortality rate

(# deaths/# people)

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Life Expectancy

(Age-Adjusted Mortality Rate)

Japan: 85

USA: 77

Throughout the semester:

You will learn how demographers